EMAN WALY

2D & 3D MEDICAL VISUALIZER | MEDICAL ILLUSTRATOR | INTERACTIVE MEDICAL CONTENT DEVELOPER

Egypt • Open to international remote opportunities
Email: emanwaly1@gmail.com | Portfolio: emanwaly1.github.io

PROFESSIONAL SUMMARY

Veterinarian and multidisciplinary medical visualizer with approximately 10 years of experience in drawing, illustration, and animation. Combines a veterinary medical background with artistic and technical skills to translate complex anatomical structures, medical conditions, and biological mechanisms into scientifically informed visual experiences. Experienced in reference-based anatomical sculpting, 3D medical modeling, scientific illustration, 2D and 3D animation, and interactive medical education. Able to develop visual projects from research and concept development through drawing, modeling, animation, and interactive implementation.

CORE EXPERTISE

- **Medical & Scientific Visualization:** Human and veterinary anatomy, anatomical modeling, pathology visualization, medical concept development.
- **3D Medical Art:** Anatomical sculpting, 3D modeling, materials, texturing, presentation.
- **Animation & Illustration:** Medical animation, frame-by-frame 2D animation, morphing transitions, scientific illustration, visual storytelling, art direction.
- **Interactive Experiences:** Interactive 3D models, clickable anatomical structures, educational quizzes, surgical learning experiences, web-based 3D visualization.

SELECTED MEDICAL VISUALIZATION PROJECTS

Human Brain — Anatomical 3D Visualization

ZBrush • Blender • Anatomical Research • Interactive 3D

- Developed a detailed human brain model through reference-based anatomical sculpting, using MRI imagery and anatomical references.
- Sculpted and organized major anatomical structures, including cerebral components, gyri, and midbrain structures.
- Modeled cerebral arteries and dural venous structures, and prepared the model for an interactive labeled 360-degree presentation with narration.

Blood Flow & Sickle Cell Disease

Blender Geometry Nodes • 3D Animation • 2D Frame-by-Frame Animation • Interactive Learning

- Created a 3D visualization comparing normal red blood cell movement with abnormal blood flow associated with sickle cell disease.
- Used Blender Geometry Nodes to control red blood cell movement and visualize cell deformation, crowding, and impaired blood flow.
- Produced a hand-drawn frame-by-frame animation explaining inheritance, hemoglobin mutation, polymerization, sickling, and vaso-occlusion.
- Developed interactive educational elements for exploration and comparison.

Human Heart & Cardiovascular Disease

3D Anatomical Modeling • Medical Animation • Interactive Visualization

- Sculpted detailed heart anatomy, including external structures and internal cross-sections, and created interactive anatomical exploration.
- Developed educational visualizations explaining ventricular septal defect (VSD), blood-flow direction, and cardiac ischemia.

Veterinary Surgical Visualization

Scientific Illustration • Interactive Surgical Learning

- Created 2D medical illustrations explaining veterinary surgical procedures, including cherry eye and entropion.
- Developed interactive surgical learning experiences and labeled veterinary anatomy content.

PROFESSIONAL EXPERIENCE

Independent Animation Production — Ahmed Al Nafees

Director • Illustrator • Animator

- Directed and executed an animated music video for nasheed artist Ahmed Al Nafees.
- Handled the complete visual production, including direction, drawing, and animation, from visual concept through final video.

Freelance Illustrator — Hindawi Foundation | 2026

Children's Publishing / Educational & Literary Content

- Produced illustrations for children's educational and literary content.
- Developed recurring characters and sequential scenes while maintaining visual consistency.
- Translated written narratives into engaging visual storytelling and delivered artwork through sketch development, revisions, and final illustration.

Background Artist — Studio Nady (Animation Studio)

Background Artist

- Worked as a background artist, creating background artwork for animated productions.

Freelance Illustrator & Animator — Independent and International Client Projects

Freelance Artist

- Created illustrations and visual narratives for educational and children's content, and produced 2D artwork and animation for remote client projects.
- Developed artwork from initial sketches through clean-up, coloring, and final production.

SELECTED WORK LINKS

- **Animated Music Video — Ahmed Al Nafees:**
https://www.youtube.com/watch?v=naDkFgyIBCo&list=RDnaDkFgyIBCo&start_radio=1
- **Children's Story Illustration — Hindawi Foundation:**
https://www.booktime.org/en/books/23074340
- **Medical Visualization Portfolio:** https://emanwaly1.github.io/

TECHNICAL SKILLS

- **3D & Sculpting:** Blender, ZBrush, Substance 3D Painter
- **2D Illustration & Design:** Adobe Photoshop, Adobe Illustrator
- **Animation & Video:** Blender Animation, Adobe After Effects, Adobe Premiere Pro, Moho
- **Interactive & Web 3D:** Three.js, JavaScript, HTML, CSS, Unity, Spline

- **3D Workflows:** Geometry Nodes, anatomical sculpting, materials, texturing, GLB export and optimization, interactive model preparation.

EDUCATION

Bachelor's Degree in Veterinary Medicine — Cairo University, Egypt (2021)

Information Technology Institute (ITI), Egypt — Software develop Fundamentals Diploma | Full Stack MERN Diploma (2023)

ADDITIONAL INFORMATION

Approximately 10 years of drawing, illustration, and animation experience. Open to paid full-time, part-time, contract, freelance, and project-based opportunities. Available for international remote collaboration.